In [1]:
```
import pandas as pd
students = pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",
                         sheet_name="students")
marks = pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",
                      sheet_name="marks")
contact = pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",
                        sheet_name="contact")
```

In [2]:
```
#1.INNER MERGE
#return only  matching Students_IDs from both DataFrame
inner_df = pd.merge(
    students,
    marks,
    on = "Student_ID",
    how ="inner"
)
print(inner_df.head())
print(inner_df.shape)
```

```
   Student_ID       Name Department  Maths  Python
0        1001  Student_1        CSE     61      56
1        1002  Student_2         IT     62      57
2        1003  Student_3        ECE     63      58
3        1004  Student_4        EEE     64      59
4        1005  Student_5       MECH     65      60
(45, 5)
```

In [3]:
```
#2.Left Merge
#Keeps all records from the left DataFrame(Students)
left_df = pd.merge(
    students,
    marks,
    on = "Student_ID",
    how ="left"
)
 06print(left_df.tail())
print(left_df.shape)
```

```
    Student_ID        Name Department  Maths  Python
45        1046  Student_46        CSE    NaN     NaN
46        1047  Student_47         IT    NaN     NaN
47        1048  Student_48        ECE    NaN     NaN
48        1049  Student_49        EEE    NaN     NaN
49        1050  Student_50       MECH    NaN     NaN
(50, 5)
```

In [5]:
```
#3.Right Merge
#Keeps all records from the Right DataFrame(Students)
Right_df = pd.merge(
    students,
    marks,
    on = "Student_ID",
    how ="left"
)
```

```
print(Right_df.tail())
print(Right_df.shape)
```

```
    Student_ID        Name Department  Maths  Python
45        1046  Student_46        CSE    NaN     NaN
46        1047  Student_47         IT    NaN     NaN
47        1048  Student_48        ECE    NaN     NaN
48        1049  Student_49        EEE    NaN     NaN
49        1050  Student_50       MECH    NaN     NaN
(50, 5)
```

In [6]:
```
#4.Outer Join
#keeps everything from both DataFrame
outer_df = pd.merge(
    students,
    marks,
    on = "Student_ID",
    how ="outer"
)
print(outer_df.tail())
print(outer_df.shape)

#5.MERGE MULTIPLS TABLES
#Combine students + marks + contact
student_complete = pd.merge(
    students,
    marks,
    on = "Student_ID",
    how ="left"
)
student_complete = pd.merge(
    student_complete,
    contact,
    on = "Student_ID",
    how ="left"
)
print(student_complete.head())
```

```
    Student_ID Name Department  Maths  Python
50        2001  NaN        NaN   75.0    80.0
51        2002  NaN        NaN   75.0    80.0
52        2003  NaN        NaN   75.0    80.0
53        2004  NaN        NaN   75.0    80.0
54        2005  NaN        NaN   75.0    80.0
(55, 5)
   Student_ID       Name Department  Maths  Python                  Email
0        1001  Student_1        CSE   61.0    56.0  student1@college.edu
1        1002  Student_2         IT   62.0    57.0  student2@college.edu
2        1003  Student_3        ECE   63.0    58.0  student3@college.edu
3        1004  Student_4        EEE   64.0    59.0  student4@college.edu
4        1005  Student_5       MECH   65.0    60.0  student5@college.edu
```

In [7]:
```
#6.Concat Row-Wise
#Split students into two parts
part1 = students.iloc[:25]
part2 = students.iloc[25:]
```

```
combined_rows = pd.concat([part1,part2])
print(combined_rows.shape)
```

```
(50, 3)
```

In [8]:
```
#7 Concat Column-Wise
student_basic = students[['Student_ID','Name']]
student_marks = marks[['Python']]
combined_cols = pd.concat(
    [student_basic,student_marks],
    axis=1
)
print(combined_cols.head())
```

```
   Student_ID       Name  Python
0        1001  Student_1      56
1        1002  Student_2      57
2        1003  Student_3      58
3        1004  Student_4      59
4        1005  Student_5      60
```

In [9]:
```
#8.JOIN - merge data
#join() works using indexes.
students_index = students.set_index("Student_ID")
contact_index = contact.set_index("Student_ID")
joined_df = students_index.join(contact_index)
print(joined_df.tail(10))
```

```
                  Name Department                  Email
Student_ID
1041        Student_41        CSE  student41@college.edu
1042        Student_42         IT  student42@college.edu
1043        Student_43        ECE  student43@college.edu
1044        Student_44        EEE  student44@college.edu
1045        Student_45       MECH  student45@college.edu
1046        Student_46        CSE  student46@college.edu
1047        Student_47         IT  student47@college.edu
1048        Student_48        ECE  student48@college.edu
1049        Student_49        EEE  student49@college.edu
1050        Student_50       MECH  student50@college.edu
```

In [ ]: